Inside the Biophoton Core

Subdivision of the core community into research strands.

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METHOD	Leiden clustering at resolution 3.0 on the coupling subgraph.

Summary

The true biophoton/UPE core is the sub-cluster where the seeds and core authors (Cifra, Popp, Van Wijk, Kobayashi, Pospíšil, Musumeci, Scordino) concentrate; the biofield/EEG/quantum-biology strands split into their own sub-clusters.

Sub-cluster 1: 335 works, 44 seeds, median 2017 (UPE core strand)

- **Topics:** Biofield Effects and Biophysics, Skin Protection and Aging, bioluminescence and chemiluminescence research, Metabolomics and Mass Spectrometry Studies
- **Authors:** Roeland Van Wijk, Michal Cifra, Eduard P.A. Van Wijk, Eduard van Wijk, Masaki Kobayashi, J. van der Greef
- rep: Ultra-weak photon emission from biological samples: Definition, mechanisms, prop (2014.0, 269c)
- rep: Vitamin E is essential for the tolerance of Arabidopsis thaliana to metal (2007.0, 186c)
- rep: In vitro electroporation detection methods: An overview (2017.0, 183c)

Sub-cluster 9: 205 works, 14 seeds, median 2011 (UPE core strand)

- **Topics:** Biofield Effects and Biophysics, Photoreceptor and optogenetics research, Plant and Biological Electrophysiology Studies, bioluminescence and chemiluminescence research
- **Authors:** Max Myakishev-Rempel, Ilya Volodyaev, Ivan Savelev, Daniel Fels, E. V. Naumova, Michal Cifra
- rep: Eukaryotic Cells and their Cell Bodies: Cell Theory Revised (2004.0, 146c)
- rep: Cellular Communication through Light (2009.0, 139c)
- rep: When microbial conversations get physical (2011.0, 123c)

Sub-cluster 4: 247 works, 12 seeds, median 2011 (UPE core strand)

- **Topics:** Photosynthetic Processes and Mechanisms, Biofield Effects and Biophysics, bioluminescence and chemiluminescence research, Photoreceptor and optogenetics research
- **Authors:** Agata Scordino, Francesco Musumeci, Rosaria Grasso, M. Gulino, S. Tudisco, A. Triglia
- rep: Monitoring and screening plant populations with combined thermal and chlorophyll (2006.0, 241c)

- rep: Biophoton emission of the human body (1997.0, 164c)
- rep: Ultraweak photon emission of human skin in vivo: Influence of topically applied (1999.0, 81c)

- **Sub-cluster 8: 214 works, 9 seeds, median 2014**
  - **Topics:** Biofield Effects and Biophysics, Paranormal Experiences and Beliefs, Functional Brain Connectivity Studies, Electromagnetic Fields and Biological Effects
  - **Authors:** Michael A. Persinger, Blake T. Dotta, Nirosha J. Murugan, Lukasz M. Karbowski, Nicolas Rouleau, Kevin S. Saroka
  - rep: Inhibition of Cancer Cell Growth by Exposure to a Specific Time-Varying Electrom (2015.0, 93c)
  - rep: Increased photon emission from the head while imagining light in the dark is cor (2012.0, 91c)
  - rep: Infrasound, human health, and adaptation: an integrative overview of recondite h (2013.0, 58c)

- **Sub-cluster 2: 288 works, 7 seeds, median 2002 (UPE core strand)**
  - **Topics:** Biofield Effects and Biophysics, bioluminescence and chemiluminescence research, Photoreceptor and optogenetics research, Plant responses to water stress
  - **Authors:** Cristiano M. Gallep, Setsuko Komatsu, Kimihiko Kato, Hiroyuki Iyozumi, Hugo J. Niggli, F. A. Popp
  - rep: In vivo imaging of spontaneous ultraweak photon emission from a rat's brain corr (1999.0, 188c)
  - rep: Biophoton emission (1984.0, 172c)
  - rep: Biophotons: ultraweak light emission from living systems (1997.0, 116c)

- **Sub-cluster 0: 474 works, 7 seeds, median 1991 (UPE core strand)**
  - **Topics:** bioluminescence and chemiluminescence research, Photodynamic Therapy Research Studies, Photoreceptor and optogenetics research, Free Radicals and Antioxidants
  - **Authors:** Roeland Van Wijk, Eduard P.A. Van Wijk, Paul S. Francis, Neil W. Barnett, Huiying Gao, Li Mao
  - rep: Singlet Molecular Oxygen Reactions with Nucleic Acids, Lipids, and Proteins (2019.0, 638c)
  - rep: Singlet molecular oxygen generated by biological hydroperoxides (2014.0, 146c)
  - rep: Free radicals and low-level photon emission in human pathogenesis: state of the (2008.0, 105c)

- **Sub-cluster 15: 155 works, 6 seeds, median 2021 (UPE core strand)**
  - **Topics:** Biofield Effects and Biophysics, Photoreceptor and optogenetics research, Neural dynamics and brain function, Laser Applications in Dentistry and Medicine
  - **Authors:** Jiapei Dai, Christoph Simon, Zhuo Wang, Vahid Salari, Paolo Grigolini, Jaimie Hoh Kam
  - rep: Optical windows for head tissues in near-infrared and short-wave infrared region (2018.0, 204c)
  - rep: A primary model of THz and far-infrared signal generation and conduction in neur (2019.0, 198c)
  - rep: Possible existence of optical communication channels in the brain (2016.0, 101c)

- **Sub-cluster 19: 114 works, 5 seeds, median 2005 (UPE core strand)**
  - **Topics:** Biofield Effects and Biophysics, Photoreceptor and optogenetics research, Quantum optics and atomic interactions, Paranormal Experiences and Beliefs
  - **Authors:** Fritz-Albert Popp, F. A. Popp, L. V. Beloussov, Jiin-Ju Chang, Kwang-Sup Soh, R. P. Bajpai
  - rep: Energy Flow in Proteins (2008.0, 316c)
  - rep: Hyperbolic relaxation as a sufficient condition of a fully coherent ergodic fiel (1993.0, 79c)
  - rep: Biophotons, coherence and photocount statistics: A critical review (2015.0, 53c)
- **Sub-cluster 5: 241 works, 5 seeds, median 2014**
  - **Topics:** Biofield Effects and Biophysics, Chemical and Physical Studies, Complementary and Alternative Medicine Studies, Spectroscopy and Quantum Chemical Studies
  - **Authors:** Pierre Madl, Larissa Brizhik, В.Л. Воейков, Alberto Tedeschi, Paolo Renati, Luigi Maxmilian Caligiuri
  - rep: Water Dynamics at the Root of Metamorphosis in Living Organisms (2010.0, 160c)
  - rep: Electromagnetic Biostimulation of Living Cultures for Biotechnology, Biofuel and (2009.0, 151c)
  - rep: High-dilution effects revisited. 2. Pharmacodynamic mechanisms (2014.0, 80c)
- **Sub-cluster 18: 116 works, 3 seeds, median 2013 (UPE core strand)**
  - **Topics:** Biofield Effects and Biophysics, Photoreceptor and optogenetics research, Retinal Development and Disorders, Visual perception and processing mechanisms
  - **Authors:** István Bókkon, Vahid Salari, Jack A. Tuszyński, Jiapei Dai, István Antal, Felix Scholkmann
  - rep: Single-photon detection by rod cells of the retina (1998.0, 271c)
  - rep: RETINAL PROCESSING NEAR ABSOLUTE THRESHOLD: From Behavior to Mechanism (2004.0, 212c)
  - rep: Imaging of Ultraweak Spontaneous Photon Emission from Human Body Displaying Diur (2009.0, 173c)
- **Sub-cluster 11: 186 works, 2 seeds, median 2004**
  - **Topics:** Electromagnetic Fields and Biological Effects, Biofield Effects and Biophysics, Magnetic and Electromagnetic Effects, Microbial Inactivation Methods
  - **Authors:** Michal Cifra, Jeremy Z. Fields, Ashkan Farhadi, Shou-Qing Ni, Zhibin Wang, Shakeel Ahmad
  - rep: Electromagnetic cellular interactions (2010.0, 332c)
  - rep: Life Rhythm as a Symphony of Oscillatory Patterns: Electromagnetic Energy and So (2014.0, 60c)
  - rep: Evidence for non-chemical, non-electrical intercellular signaling in intestinal (2007.0, 51c)
- **Sub-cluster 3: 273 works, 1 seeds, median 1994 (UPE core strand)**
  - **Topics:** Biofield Effects and Biophysics, Electromagnetic Fields and Biological Effects, Terahertz technology and applications, Spectroscopy and Quantum Chemical Studies

- **Authors:** Jordane Preto, Marco Pettini, Michal Cifra, J. Pokorný, Jiří Hašek, F Jelínek
- rep: Terahertz radiation induces non-thermal structural changes associated with Fröhl (2015.0, 87c)
- rep: ELECTROMAGNETIC ACTIVITY OF YEAST CELLS IN THE M PHASE (2001.0, 77c)
- rep: Possible role of electrodynamic interactions in long-distance biomolecular recog (2015.0, 64c)

Sub-cluster 10: 199 works, 1 seeds, median 2013 (UPE core strand)

- **Topics:** Microtubule and mitosis dynamics, Biofield Effects and Biophysics, Electromagnetic Fields and Biological Effects, Photoreceptor and optogenetics research
- **Authors:** Jiří Pokorný, J. Pokorný, Jan Vrba, Michal Cifra, Ondřej Kučera, Jack A. Tuszyński
- rep: Nanowired Bioelectric Interfaces (2019.0, 142c)
- rep: An Overview of Sub-Cellular Mechanisms Involved in the Action of TTFields (2016.0, 94c)
- rep: Cancer physics: diagnostics based on damped cellular elastoelectrical vibrations (2011.0, 67c)

Sub-cluster 6: 239 works, 1 seeds, median 2014

- **Topics:** Spectroscopy and Quantum Chemical Studies, Photoreceptor and optogenetics research, Photosynthetic Processes and Mechanisms, Biofield Effects and Biophysics
- **Authors:** Graham R. Fleming, Gregory D. Scholes, Gregory S. Engel, Paul Brumer, Martin B. Plenio, Susana F. Huelga
- rep: Evidence for wavelike energy transfer through quantum coherence in photosyntheti (2007.0, 3335c)
- rep: Lessons from nature about solar light harvesting (2011.0, 1924c)
- rep: Coherently wired light-harvesting in photosynthetic marine algae at ambient temp (2010.0, 1663c)

Sub-cluster 7: 237 works, 1 seeds, median 2007

- **Topics:** Quantum Mechanics and Applications, Biofield Effects and Biophysics, Neural dynamics and brain function, Photoreceptor and optogenetics research
- **Authors:** Stuart R. Hameroff, Jack A. Tuszyński, Roger Penrose, Travis J. A. Craddock, Sara Diani, Vahid Salari
- rep: Consciousness in the universe (2013.0, 937c)
- rep: Consciousness in the universe: a review of the 'Orch OR' theory. (2014.0, 384c)
- rep: Conscious Events as Orchestrated Space-Time Selections (2007.0, 202c)

Sub-cluster 12: 182 works, 1 seeds, median 2014

- **Topics:** Laser Applications in Dentistry and Medicine, Photoreceptor and optogenetics research, Nanoplatforms for cancer theranostics, bioluminescence and chemiluminescence research
- **Authors:** Gerd Hoffmann, Michael R. Hamblin, Kanyi Pu, Hong Bae Kim, Ku Youn Baik, Pill-Hoon Choung
- rep: Near-infrared fluorophores for biomedical imaging (2017.0, 2935c)
- rep: Fluorescence imaging in vivo: recent advances (2007.0, 778c)

- rep: Biological effects and medical applications of infrared radiation (2017.0, 474c)

- **Sub-cluster 13: 170 works, 1 seeds, median 2009**
  - **Topics:** Neural dynamics and brain function, EEG and Brain-Computer Interfaces, Functional Brain Connectivity Studies, Biofield Effects and Biophysics
  - **Authors:** Joachim Keppler, L. A. Cacha, Roman R. Poznański, А. Б. Бурлаков, O. V. Burlakova, В. А. Голиченков
  - rep: The ADaptation and Anticipation Model (ADAM) of sensorimotor synchronization (2013.0, 580c)
  - rep: The “conscious pilot”, dendritic synchrony moves through the brain to mediate con (2009.0, 110c)
  - rep: Neurophysiological correlates of error correction in sensorimotor-synchronizatio (2003.0, 73c)

- **Sub-cluster 14: 168 works, 1 seeds, median 2012**
  - **Topics:** Quantum Information and Cryptography, Quantum optics and atomic interactions, Random lasers and scattering media, Quantum Mechanics and Applications
  - **Authors:** Lingyan Shi, R. R. Alfano, Sandra Mamani, Daniel A. Nolan, Shaul Mukamel, Daniel Fels
  - rep: Quantum metrology and its application in biology (2015.0, 482c)
  - rep: Nonlinear optical signals and spectroscopy with quantum light (2016.0, 351c)
  - rep: Multidimensional pump-probe spectroscopy with entangled twin-photon states (2009.0, 69c)

- **Sub-cluster 16: 137 works, 1 seeds, median 2022**
  - **Topics:** Photoreceptor and optogenetics research, Biofield Effects and Biophysics, Quantum Mechanics and Applications, Microtubule and mitosis dynamics
  - **Authors:** Jack A. Tuszyński, Michael Wiest, Josh Roeloffs, Travis J. A. Craddock, Lucio Tonello, Massimo Cocchi
  - rep: Microtubules as a potential platform for energy transfer in biological systems: (2022.0, 44c)
  - rep: Photon Entanglement Through Brain Tissue (2016.0, 43c)
  - rep: The quantum mitochondrion and optimal health (2016.0, 42c)

- **Sub-cluster 20: 110 works, 1 seeds, median 2017**
  - **Topics:** bioluminescence and chemiluminescence research, Bacterial biofilms and quorum sensing, Cephalopods and Marine Biology, Protist diversity and phylogeny
  - **Authors:** Maya Menashe, Liora Rahmani, Stefan Schramm, Dieter G. Weiss, Youri Timsit, Magali Lescot
  - rep: Bioluminescence: a versatile technique for imaging cellular and molecular featur (2013.0, 118c)
  - rep: Bioluminescence: The Vibrant Glow of Nature and its Chemical Mechanisms (2024.0, 50c)
  - rep: Bioluminescence and Photoreception in Unicellular Organisms: Light-Signalling in (2021.0, 24c)

- **Sub-cluster 25: 88 works, 1 seeds, median 2018**
  - **Topics:** Biofield Effects and Biophysics, Neural dynamics and brain function, Quantum Mechanics and Applications, Spectroscopy and Quantum Chemical Studies
  - **Authors:** Akihiro Nishiyama, Shigenori Tanaka, Giuseppe Vitiello, Jack A. Tuszyński, Carlo Dal Lin, Sabino Iliceto
  - rep: Electromagnetic field and spontaneous symmetry breaking in biological matter (1986.0, 137c)
  - rep: Sounds Stimulation on In Vitro HL1 Cells: A Pilot Study and a Theoretical Physic (2020.0, 28c)
  - rep: Evanescent (tunneling) photon and cellular vision' (1997.0, 27c)
- **Sub-cluster 28: 48 works, 1 seeds, median 2014**
  - **Topics:** Effects of Radiation Exposure, Extracellular vesicles in disease, Biofield Effects and Biophysics, Cancer Cells and Metastasis
  - **Authors:** Carmel Mothersill, Colin Seymour, Andrew J. Rainbow, Fiona E. McNeill, M.S. Le, Michelle Le
  - rep: Exosomes are released by bystander cells exposed to radiation-induced biophoton (2017.0, 90c)
  - rep: Modulation of oxidative phosphorylation (OXPHOS) by radiation- induced biophoton (2018.0, 47c)
  - rep: Quantum Biology and the Potential Role of Entanglement and Tunneling in Non-Targ (2023.0, 38c)
- **Sub-cluster 26: 73 works, 0 seeds, median 2014 (UPE core strand)**
  - **Topics:** Chemical Reactions and Isotopes, Biofield Effects and Biophysics, Photoreceptor and optogenetics research, Hydrogen's biological and therapeutic effects
  - **Authors:** Ignat Ignatov, Oleg Mosin, Christos Drossinakis, Hugo J. Niggli, Chavdar Stoyanov, Reneta Toshkova
  - rep: Evaluating Possible Methods and Approaches for Registering of Electromagnetic Wa (2014.0, 66c)
  - rep: Biophysical Fields. Color Coronal Spectral Analysis. Registration with Water Spe (2014.0, 28c)
  - rep: Registration of Electromagnetic Waves Emitted from the Human Body (2014.0, 27c)
- **Sub-cluster 22: 106 works, 0 seeds, median 2016**
  - **Topics:** Biofield Effects and Biophysics, Acupuncture Treatment Research Studies, Neonatal Respiratory Health Research, Electromagnetic Fields and Biological Effects
  - **Authors:** Ганна Володимирівна Невойт, Maksim Potyazhenko, Alfonsas Vainoras, О. П. Мінцер, Gediminas Jaruševičius, Inga Arūnė Bumblytė
  - rep: Understanding Traditional Chinese Medicine Therapeutics: An Overview of the Basi (2021.0, 295c)
  - rep: Characterization of DNA-containing Granules Flowing Through the Meridian-like Sy (2006.0, 34c)

- rep: EVALUATION OF THE HUMAN BIOELECTROMAGNETIC FIELD IN MEDICINE: THE DEVELOPMENT OF (2019.0, 30c)

- **Sub-cluster 24: 90 works, 0 seeds, median 2022**
  - **Topics:** Plant and Biological Electrophysiology Studies, Origins and Evolution of Life, Photoreceptor and optogenetics research, Molecular Communication and Nanonetworks
  - **Authors:** William B. Miller, František Baluška, Arthur S. Reber, Predrag Slijepčević, John S. Torday, Bruno Bordoni
  - rep: Biological information systems: Evolution as cognition-based information managem (2017.0, 98c)
  - rep: Biomolecular Basis of Cellular Consciousness via Subcellular Nanobrains (2021.0, 96c)
  - rep: Towards the Idea of Molecular Brains (2021.0, 68c)

- **Sub-cluster 17: 132 works, 0 seeds, median 2014**
  - **Topics:** Biofield Effects and Biophysics, Complementary and Alternative Medicine Studies, Cellular Mechanics and Interactions, Paranormal Experiences and Beliefs
  - **Authors:** Rick Sá, Gaétan Chevalier, Patrizio Tressoldi, Luciano Pederzoli, John G. Kruth, Marie Grace Brook
  - rep: An Overview of Biofield Devices (2015.0, 47c)
  - rep: Biofield Science: Current physics perspectives (2015.0, 46c)
  - rep: Extended Network Generalized Entanglement Theory: Therapeutic Mechanisms, Empiri (2003.0, 41c)

- **Sub-cluster 21: 110 works, 0 seeds, median 2019**
  - **Topics:** Electromagnetic Fields and Biological Effects, Photoreceptor and optogenetics research, Magnetic and Electromagnetic Effects, Light effects on plants
  - **Authors:** Hadi Zadeh-Haghighi, Christoph Simon, Betony Adams, Francesco Petruccione, Alistair V.W. Nunn, Geoffrey W. Guy
  - rep: The Quantum Biology of Reactive Oxygen Species Partitioning Impacts Cellular Bio (2016.0, 136c)
  - rep: Radical pairs may explain reactive oxygen species-mediated effects of hypomagnet (2022.0, 39c)
  - rep: Hypomagnetic Conditions and Their Biological Action (Review) (2023.0, 27c)

- **Sub-cluster 23: 93 works, 0 seeds, median 2012**
  - **Topics:** Origins and Evolution of Life, Gene Regulatory Network Analysis, Plant and Biological Electrophysiology Studies, Quantum Mechanics and Applications
  - **Authors:** Abir U. Igamberdiev, Richard Gordon, Pierre Madl, Hanno Stutz, Ione Hunt von Herbing, Lucio Tonello
  - rep: Toward understanding the emergence of life: A dual function of the system of nuc (2023.0, 34c)

- rep: Biomechanical and coherent phenomena in morphogenetic relaxation processes (2012.0, 25c)
- rep: Reflexive neural circuits and the origin of language and music codes (2024.0, 23c)

- **Sub-cluster 27: 65 works, 0 seeds, median 2017**
  - **Topics:** Myofascial pain diagnosis and treatment, Photoreceptor and optogenetics research, Planarian Biology and Electrostimulation, Mesenchymal stem cell research
  - **Authors:** Carlo Ventura, W Jaroß, Irena Ćosić, Drasko Cosic, Rostislav Bychkov, Magdalena Juhaszova
  - rep: Synchronized Cardiac Impulses Emerge From Heterogeneous Local Calcium Signals Wi (2020.0, 93c)
  - rep: Tissue Regeneration without Stem Cell Transplantation: Self-Healing Potential fr (2018.0, 30c)
  - rep: Environmental Light and Its Relationship with Electromagnetic Resonances of Biom (2016.0, 23c)

- **Sub-cluster 29: 34 works, 0 seeds, median 2017**
  - **Topics:** Superconducting and THz Device Technology, GaN-based semiconductor devices and materials, Adaptive optics and wavefront sensing, Electronic and Structural Properties of Oxides
  - **Authors:** Benjamin A. Mazin, Seth R. Meeker, K. O'Brien, Paul Szypryt, Alexander B. Walter, B. Bumble
  - rep: Lumped Element Kinetic Inductance Detectors (2008.0, 275c)
  - rep: Titanium nitride films for ultrasensitive microresonator detectors (2010.0, 245c)
  - rep: ARCONS: A 2024 Pixel Optical through Near-IR Cryogenic Imaging Spectrophotometer (2013.0, 172c)

- **Sub-cluster 30: 19 works, 0 seeds, median 2022**
  - **Topics:** Photoreceptor and optogenetics research, Mechanical and Optical Resonators, Quantum, superfluid, helium dynamics, Quantum Mechanics and Applications
  - **Authors:** Jack A. Tuszyński, R Poznanski
  - rep: The activity of information in biomolecular systems: a fundamental explanation o (2022.0, 18c)
  - rep: The archetypal molecular patterns of conscious experience are quantum analogs. (2022.0, 9c)
  - rep: nan (2022.0, 0c)